CHEMISTRY | UNIT 02 | LESSON 05

Periodic Trends and Chemical Properties

Checkpoint Submission

Student name:		Date:	
MLA standard:	MLA.CHEM.ATO.03	Submission:	Checkpoint Submission

Directions

Complete independently. Use the Period 3 data and the periodic-trend rules taught in the lesson. Submit this PDF to U2 L5 Checkpoint Submission.

Part A - Period 3 evidence

Element	Atomic #	Atomic radius (pm)	1st ionization energy (kJ/mol)	Electronegativity
Na	11	186	496	0.93
Mg	12	160	738	1.31
Al	13	143	578	1.61
Si	14	118	787	1.90
P	15	110	1012	2.19
S	16	103	1000	2.58
Cl	17	99	1251	3.16

Part B - Apply the trends

1. Rank Na, Al, and Cl from largest to smallest atomic radius. Cite the table values.

2. Rank Na, Al, and Cl from lowest to highest first ionization energy. Cite the table values.

3. A student claims that every step across a period must increase ionization energy. Use Mg and Al or P and S to evaluate the claim.

4. Compare Na and Cl. Explain which is more metallic and which attracts shared electrons more strongly.

5. Predict whether potassium or sodium has the larger atomic radius. Explain using period, shielding, and occupied energy levels.

Mastery check

Use accurate vocabulary: atomic radius, ionization energy, electronegativity, nuclear charge, shielding, period, and group.